

Phase Diversity Adaptive Optics for Multiphoton Microscopy

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1 Forward Model

Let $\mathbf{u}$ be the coordinates of the back focal plane (which we treat as two-dimensional), and $\mathbf{x}$ be the coordinates of the front focal plane (which we treat as three-dimensional). We denote a 3D stack of images as a map from $\mathbb{R}^3$ to the nonnegative real line:

$$d : \mathbb{R}^3 \rightarrow [0, \infty) \quad (1)$$

The image stack is generated by convolving a 3D PSF $s_k : \mathbb{R}^3 \rightarrow [0, \infty)$ with the underlying 3D object $f : \mathbb{R}^3 \rightarrow [0, \infty)$. It is often convenient to consider the individual images separately. We will label the images with the index k , and write:

$$d_k(x, y) = \sum_{i=1}^P f(x, y, z_i) \otimes s_k(x, y, z_i - z_k) \quad (2)$$

where z_k is the focal plane of the image, and $\{z_i\}$ are the P planes around this focal plane.

The PSF is generated by taking the pupil function $H_k : \mathbb{R}^2 \rightarrow \mathbb{C}$, applying defocus according to the z -component of $\mathbf{x}$, performing the inverse Fourier transform, and raising the squared magnitude of the result to the N th power:

$$s_k(\mathbf{x}) = |h_k(\mathbf{x})|^{2N} \quad h_k(\mathbf{x}) = \mathcal{F}^{-1}\{H_k(\mathbf{u})e^{iz\gamma(\mathbf{u})}\} \quad (3)$$

The extra defocus is what sets this model apart from 2D. We calculate it with:

$$\gamma(\mathbf{u}) = \frac{2\pi n_m}{\lambda} \sqrt{1 - \left(\frac{\lambda \|\mathbf{u}\|}{n_m}\right)^2} \quad (4)$$

where n_m is the refractive index and λ is the vacuum wavelength. Furthermore, the pupil function is generated by a pupil mask/apodization $P(\mathbf{u})$, an underlying phase aberration $\phi(\mathbf{u})$ and a known diversity phase $\theta_k(\mathbf{u})$. In all, we write this as:

$$H_k(\mathbf{u}) = |P(\mathbf{u})|e^{i(\phi(\mathbf{u})+\theta_k(\mathbf{u}))} \quad (5)$$

Sometimes, we will expand $\phi(\mathbf{u})$ in a basis $\{Z_i\}$. In this case, we will use the vector $\boldsymbol{\alpha}$ to describe the coefficients of ϕ in this basis.

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2 Cost Functionals

Let $\{d_k\}$ be the set of acquired images, with known diversity phases $\{\theta_k\}$. The goal is to estimate the underlying aberration $\phi(\mathbf{u})$ and the underlying sample $f(\mathbf{x})$ to minimize a cost functional $\mathcal{L}\{f(\mathbf{x}), \boldsymbol{\alpha}\}$. In keeping with the literature, we will treat two cases: Gaussian and Poisson noise. In the Gaussian case, we seek to minimize the least-squares objective:

$$\mathcal{L}_g\{f, \boldsymbol{\alpha}\} = \sum_{k=1}^K \sum_{(x,y)} [d_k(x, y) - \mu_k(x, y)]^2 \quad (6)$$

where we use the shorthand:

$$\mu_k(x, y) = \sum_{i=1}^P f(x, y, z_i) \otimes s_k(x, y, z_i - z_k) \quad (7)$$

where s_k here denotes the prediction of the PSF from $\boldsymbol{\alpha}$. We can also write out the Poisson objective:

$$\mathcal{L}_p\{f, \boldsymbol{\alpha}\} = 2 \sum_{k=1}^K \sum_{(x,y)} [d_k(x, y) \ln(d_k(x, y)) - d_k(x, y) \ln(\mu_k(x, y)) - (d_k(x, y) - \mu_k(x, y))] \quad (8)$$

3 Gaussian Noise

We will first attempt to solve the problem for the case of Gaussian noise.

3.1 Object Estimation

We begin by estimating the underlying object f . By Parseval's theorem, we can rewrite the Gaussian cost functional in [Equation 6](#) in the Fourier domain:

$$\mathcal{L}_g\{F, \boldsymbol{\alpha}\} = \sum_{k=1}^K \sum_{(u_x, u_y)} \left| D_k - \sum_{i=1}^P F_{z_i} S_k(z_i - z_k) \right|^2 \quad (9)$$

where D_k denotes the Fourier transform of d_k , F_{z_i} denotes the Fourier transform of $f(x, y, z_i)$ and $S_k(z_i - z_k)$ denotes the Fourier transform of the k th simulated PSF slice at $z_i - z_k$. The notation $\mathcal{F}_{xy}$ used below denotes the same lateral Fourier transform. At each fixed spatial frequency, the unconstrained object that minimizes the cost functional will satisfy $\partial \mathcal{L}_g / \partial F_{z_j}^* = 0$ for each $j = 1, \dots, P$. We compute each of these w.r.t the complex conjugate of F_{z_j} so that the resulting expression will include F_{z_j} directly:

$$\frac{\partial \mathcal{L}_g}{\partial F_{z_j}^*} = \frac{\partial}{\partial F_{z_j}^*} \sum_{k=1}^K \sum_{(u_x, u_y)} \left| D_k - \sum_{i=1}^P F_{z_i} S_k(z_i - z_k) \right|^2 \quad (10)$$

$$= \sum_{k=1}^K \frac{\partial}{\partial F_{z_j}^*} \left[\left(D_k - \sum_{i=1}^P F_{z_i} S_k(z_i - z_k) \right) \left(D_k^* - \sum_{i=1}^P F_{z_i}^* S_k^*(z_i - z_k) \right) \right] \quad (11)$$

$$= - \sum_{k=1}^K \left[S_k^*(z_j - z_k) \left(D_k - \sum_{i=1}^P F_{z_i} S_k(z_i - z_k) \right) \right] \quad (12)$$

Rearranging, we get:

$$\sum_{i=1}^P \left[\sum_{k=1}^K S_k^*(z_j - z_k) S_k(z_i - z_k) \right] F_{z_i} = \sum_{k=1}^K S_k^*(z_j - z_k) D_k \quad (13)$$

We get P equations of this form, one for each index $j = 1, \dots, P$. Together, they form the matrix equation:

$$(\mathbf{S}^\dagger \mathbf{S}) \mathbf{F} = \mathbf{S}^\dagger \mathbf{D} \quad (14)$$

It is helpful to briefly remind ourselves of the dimensions of these matrices: $\mathbf{F}$ is a single column with one row for each equation (P). By similar logic, we can infer that $\mathbf{D}$ has one column and one row for each image (K). $\mathbf{S}$ is $K \times P$, and $\mathbf{S}^\dagger$ is $P \times K$, so, $\mathbf{S}^\dagger \mathbf{S}$ is $P \times P$. We can thus verify the dimensions of our equation are correct. Solving, we get the estimator:

$$\hat{\mathbf{F}} = (\mathbf{S}^\dagger \mathbf{S})^{-1} \mathbf{S}^\dagger \mathbf{D} \quad (15)$$

where we have assumed that $\mathbf{S}^\dagger \mathbf{S}$ is invertible. This means that all of the columns of $\mathbf{S}$ must be linearly independent. Since $\mathbf{S}$ is $K \times P$, it must be the case that $K \geq P$; otherwise this is impossible. As Kner points out, this tells us that we must have at least P measurements to estimate the image at P planes.

Sometimes, we would like an estimate, even when we have insufficient data. We therefore introduce a fixed regularization parameter $\rho > 0$ as follows:

$$\hat{\mathbf{F}} = (\mathbf{S}^\dagger \mathbf{S} + \rho I_P)^{-1} \mathbf{S}^\dagger \mathbf{D} \quad (16)$$

which will allow us to solve for the inverse even if $\mathbf{S}$ has insufficient rank. This is equivalent to placing a squared L^2 norm penalty on $\mathbf{F}$ in the original cost function [Equation 9](#).

3.2 Aberration Estimation

Next, we will discuss the process of estimating the aberration along a basis $\{Z_i\}$, with real coefficients α_i :

$$\phi(\mathbf{u}) = \sum_i \alpha_i Z_i(\mathbf{u}) \quad (17)$$

We will begin by defining the following variable, the residual between our k th estimated image (which we generate from the estimated object $\hat{\mathbf{F}}$) and the actual image D_k :

$$r_k = D_k - \sum_{i=1}^P \hat{F}_{z_i} S_k(z_i - z_k) \quad (18)$$

With $\hat{\mathbf{F}}$ plugged in, and the additional squared L^2 penalty added, we rewrite the original cost functional [Equation 9](#) as:

$$\mathcal{L}_g\{\boldsymbol{\alpha}\} = \sum_{(u_x, u_y)} \sum_{k=1}^K r_k r_k^* + \rho \sum_{(u_x, u_y)} \sum_{i=1}^P \hat{F}_{z_i} \hat{F}_{z_i}^* \quad (19)$$

To take the gradient $\nabla \mathcal{L}_g$, we begin by taking the derivative with respect to an arbitrary aberration coefficient α_m :

$$\frac{\partial \mathcal{L}_g}{\partial \alpha_m} = \sum_{(u_x, u_y)} \sum_{k=1}^K \left[r_k^* \left(\frac{\partial r_k}{\partial \alpha_m} \right) + r_k \left(\frac{\partial r_k}{\partial \alpha_m} \right)^* \right] + \rho \sum_{(u_x, u_y)} \sum_{i=1}^P \left[\hat{F}_{z_i}^* \left(\frac{\partial \hat{F}_{z_i}}{\partial \alpha_m} \right) + \hat{F}_{z_i} \left(\frac{\partial \hat{F}_{z_i}}{\partial \alpha_m} \right)^* \right] \quad (20)$$

In both terms, we recognize sums of the form $w + w^* = 2\Re(w)$. Using this identity, we rewrite:

$$\frac{\partial \mathcal{L}_g}{\partial \alpha_m} = 2\Re \left[\sum_{(u_x, u_y)} \sum_{k=1}^K r_k^* \frac{\partial r_k}{\partial \alpha_m} \right] + 2\rho \Re \left[\sum_{(u_x, u_y)} \sum_{i=1}^P \hat{F}_{z_i}^* \frac{\partial \hat{F}_{z_i}}{\partial \alpha_m} \right] \quad (21)$$

We examine the next derivative:

$$\frac{\partial r_k}{\partial \alpha_m} = \frac{\partial}{\partial \alpha_m} \left[D_k - \sum_{i=1}^P \hat{F}_{z_i} S_k(z_i - z_k) \right] = - \sum_{i=1}^P \left[\frac{\partial \hat{F}_{z_i}}{\partial \alpha_m} S_k(z_i - z_k) + \hat{F}_{z_i} \frac{\partial S_k(z_i - z_k)}{\partial \alpha_m} \right] \quad (22)$$

From here on, for compactness we denote:

$$S_{ki} = S_k(z_i - z_k) \quad (23)$$

Plugging in our results, we get:

$$\frac{\partial \mathcal{L}_g}{\partial \alpha_m} = -2\Re \left[\sum_{(u_x, u_y), k, i} r_k^* S_{ki} \frac{\partial \hat{F}_{z_i}}{\partial \alpha_m} \right] - 2\Re \left[\sum_{(u_x, u_y), k, i} r_k^* \hat{F}_{z_i} \frac{\partial S_{ki}}{\partial \alpha_m} \right] + 2\rho \Re \left[\sum_{(u_x, u_y), i} \hat{F}_{z_i}^* \frac{\partial \hat{F}_{z_i}}{\partial \alpha_m} \right] \quad (24)$$

We briefly examine the first and third terms, since they both involve the derivative of the optimized object with respect to the aberration coefficient. Inside the real-part operator, we have:

$$- \sum_{(u_x, u_y)} \sum_i \left[\sum_k r_k^* S_{ki} \right] \frac{\partial \hat{F}_{z_i}}{\partial \alpha_m} + \rho \sum_{(u_x, u_y)} \sum_i \hat{F}_{z_i}^* \frac{\partial \hat{F}_{z_i}}{\partial \alpha_m} \quad (25)$$

However, recall from [Equation 16](#) that:

$$(\mathbf{S}^\dagger \mathbf{S} + \rho I_P) \hat{\mathbf{F}} = \mathbf{S}^\dagger (\mathbf{r} + \mathbf{S} \hat{\mathbf{F}}) \quad (26)$$

which gives:

$$\rho \hat{\mathbf{F}} = \mathbf{S}^\dagger \mathbf{r} \quad (27)$$

which allows us to cancel the corresponding terms in the summation. The result is:

$$\frac{\partial \mathcal{L}_g}{\partial \alpha_m} = -2\Re \left[\sum_{(u_x, u_y), k, i} r_k^* \hat{F}_{z_i} \frac{\partial S_{ki}}{\partial \alpha_m} \right] \quad (28)$$

The next task is to evaluate the PSF derivative. Recall the forward model:

$$\frac{\partial S_{ki}}{\partial \alpha_m} = \frac{\partial}{\partial \alpha_m} \mathcal{F}\{s_k(x, y, z_i - z_k)\} = \mathcal{F}\left\{ \frac{\partial s_k(x, y, z_i - z_k)}{\partial \alpha_m} \right\} \quad (29)$$

where

$$\frac{\partial s_k(x, y, z_i - z_k)}{\partial \alpha_m} = \frac{\partial}{\partial \alpha_m} (h_k h_k^*)^N = 2N |h_k|^{2N-2} \Re \left[h_k^* \frac{\partial h_k}{\partial \alpha_m} \right] \quad (30)$$

where

$$\frac{\partial h_k}{\partial \alpha_m} = \frac{\partial}{\partial \alpha_m} \mathcal{F}^{-1}\{H_k(\mathbf{u}) e^{iz\gamma(\mathbf{u})}\} = \mathcal{F}^{-1}\left\{ i Z_m(\mathbf{u}) H_k(\mathbf{u}) e^{iz\gamma(\mathbf{u})} \right\} \quad (31)$$

where we have used that the only α_m dependence in this term is in H_k , where:

$$\frac{\partial H_k}{\partial \alpha_m} = \frac{\partial}{\partial \alpha_m} |P(\mathbf{u})| e^{i(\phi(\mathbf{u}) + \theta_k(\mathbf{u}))} = i Z_m H_k \quad (32)$$

by expanding ϕ in our basis as $\sum_j Z_j \alpha_j$. Summarizing, we have:

$$\begin{aligned} \frac{\partial \mathcal{L}_g}{\partial \alpha_m} = & -4N \Re \sum_{(u_x, u_y)} \sum_{k=1}^K \left[D_k - \sum_{i=1}^P \hat{F}_{z_i} S_k(z_i - z_k) \right]^* \\ & \times \left[\sum_{j=1}^P \hat{F}_{z_j} \mathcal{F} \left\{ |h_k(z_j - z_k)|^{2N-2} \Re \left[h_k^*(z_j - z_k) \mathcal{F}^{-1} \left\{ i Z_m H_k e^{i(z_j - z_k)\gamma} \right\} \right] \right\} \right]. \end{aligned} \quad (33)$$

I then used GPT-6 Astra to automatically compute the Hessian, which takes the following form:

$$\begin{aligned} \frac{\partial^2 \mathcal{L}_g}{\partial \alpha_m \partial \alpha_\ell} = & 8N^2 \Re \sum_{(u_x, u_y)} \sum_{k=1}^K \left[\sum_{i=1}^P \hat{F}_{z_i} \mathcal{F}_{xy} \left\{ |h_k(z_i - z_k)|^{2N-2} a_{ki,m} \right\} \right]^* \\ & \times \left[\sum_{j=1}^P \hat{F}_{z_j} \mathcal{F}_{xy} \left\{ |h_k(z_j - z_k)|^{2N-2} a_{kj,\ell} \right\} \right] \\ & - 8N(N-1) \Re \sum_{(u_x, u_y)} \sum_{k=1}^K r_k^* \sum_{j=1}^P \hat{F}_{z_j} \mathcal{F}_{xy} \left\{ |h_k(z_j - z_k)|^{2N-4} a_{kj,m} a_{kj,\ell} \right\} \\ & - 4N \Re \sum_{(u_x, u_y)} \sum_{k=1}^K r_k^* \sum_{j=1}^P \hat{F}_{z_j} \mathcal{F}_{xy} \left\{ |h_k(z_j - z_k)|^{2N-2} b_{kj,m\ell} \right\} \\ & - 2 \Re \sum_{(u_x, u_y)} \sum_{i=1}^P \sum_{j=1}^P c_{i,m}^* \left[(\mathbf{S}^\dagger \mathbf{S} + \rho I_P)^{-1} \right]_{ij} c_{j,\ell}. \end{aligned} \quad (34)$$

where:

$$\begin{aligned}
a_{ki,m} &= \text{Re} \left[h_k^*(z_i - z_k) \mathcal{F}^{-1} \left\{ i Z_m H_k e^{i(z_i - z_k)\gamma} \right\} \right], \\
b_{ki,m\ell} &= \text{Re} \left[\left(\mathcal{F}^{-1} \left\{ Z_\ell H_k e^{i(z_i - z_k)\gamma} \right\} \right)^* \mathcal{F}^{-1} \left\{ Z_m H_k e^{i(z_i - z_k)\gamma} \right\} \right. \\
&\quad \left. - h_k^*(z_i - z_k) \mathcal{F}^{-1} \left\{ Z_m Z_\ell H_k e^{i(z_i - z_k)\gamma} \right\} \right], \\
c_{j,m} &= 2N \sum_{k=1}^K \left[\left(\mathcal{F}_{xy} \left\{ |h_k(z_j - z_k)|^{2N-2} a_{kj,m} \right\} \right)^* r_k \right. \\
&\quad \left. - S_k^*(z_j - z_k) \sum_{i=1}^P \widehat{F}_{z_i} \mathcal{F}_{xy} \left\{ |h_k(z_i - z_k)|^{2N-2} a_{ki,m} \right\} \right], \\
r_k &= D_k - \sum_{i=1}^P \widehat{F}_{z_i} S_k(z_i - z_k), \\
h_k(z) &= \mathcal{F}^{-1} \left\{ H_k e^{iz\gamma} \right\}, \\
S_k(z) &= \mathcal{F}_{xy} \left\{ |h_k(z)|^{2N} \right\}, \\
[\mathbf{S}]_{ki} &= S_k(z_i - z_k), \\
\mathbf{D} &= (D_1, \dots, D_K)^\top, \\
\widehat{\mathbf{F}} &= (\widehat{F}_{z_1}, \dots, \widehat{F}_{z_P})^\top = (\mathbf{S}^\dagger \mathbf{S} + \rho I_P)^{-1} \mathbf{S}^\dagger \mathbf{D}, \\
\mathcal{L}_g(\boldsymbol{\alpha}) &= \sum_{(u_x, u_y)} \left[\sum_{k=1}^K |r_k|^2 + \rho \sum_{i=1}^P |\widehat{F}_{z_i}|^2 \right].
\end{aligned}$$